

1. Purpose and Scope

This document develops a reduced structural formalism for the Collatz map by factoring out deterministic valuation behavior into extended valuation blocks and studying the induced dynamics on reduced states (ω, d) . The objective is not to analyze the iteration step-by-step, but to isolate the arithmetic structure that remains once these deterministic block effects are compressed.

The central hope of this framework is that the reduced dynamics may reveal a more connected and intelligible state structure than is visible in the raw iteration. In particular, the goal is to understand how reduced states relate to one another, whether transitions admit further reduction, and whether the induced map exposes structural regularities that are obscured in the standard formulation.

No claim is made here regarding convergence or a resolution of the Collatz conjecture. The purpose of this note is only to define the reduced formalism clearly, establish its exact algebraic properties, and identify which additional patterns appear to merit further investigation.

2. Layered View of the Collatz Map

The reduction developed in this note is best understood as a layered compression of the ordinary Collatz dynamics rather than as a single reformulation. Each layer isolates a different kind of structure, and only the final layer introduces the genuinely reduced state space studied here.

Layer 1. Raw Collatz Iteration

At the first layer sits the ordinary Collatz map on the positive integers. This is the classical step-by-step dynamics, whose local behavior is easy to define but whose longer-range arithmetic structure is difficult to read directly from iteration alone.

Layer 2. Extended Valuation Blocks

The second layer groups together contiguous stretches of the dynamics in which the valuation behavior of $x + 1$ evolves deterministically. These extended valuation blocks compress repeated local behavior that does not need to be re-analyzed at every step. This layer is still an exact reformulation of the same dynamics, but organized around deterministic block structure rather than individual Collatz steps.

Layer 3. BlockEntry Coordinates (u, m)

The third layer records each block at its entry in canonical coordinates. Here u denotes the odd seed associated with the block entry and m records the local entry depth. These coordinates describe the block exactly and provide the natural input data for the structural step formula developed later.

Layer 4. Reduced Structural States (ω, d)

The fourth layer passes from exact block coordinates to reduced structural states. Writing $u = 3^a \omega$ with 3 not dividing ω , the reduced state is obtained by retaining the structural odd core ω together with the full combined depth $d = m + a$. This is the genuinely reduced layer: different BlockEntry coordinates may determine the same reduced state, and the induced map on these reduced states is the main object studied in this note.

The first three layers should be viewed primarily as exact organization of the dynamics: they isolate deterministic structure, choose convenient coordinates, and prepare the quotient. The fourth layer is the novel step of the framework, where part of the arithmetic data is factored out and the dynamics are studied through the induced transition on reduced states.

No claim is made here that the earlier layers are historically new in themselves. The point is instead that they provide the scaffolding needed to define the reduced structural state space cleanly and to formulate the induced map on that space.

Running Example

A small example helps illustrate how the same local trajectory appears at each layer. Consider the odd value 23. Then

$$23 + 1 = 24 = 2^3 \cdot 3,$$

so the associated block-entry data are $u = 3$ and $m = 3$, since $23 = 2^m u - 1$. In reduced form, $u = 3^1 \cdot 1$, so $a = 1$, $\omega = 1$, and $d = m + a = 4$.

Viewed through the four layers:

- At Layer 1, one follows the ordinary Collatz trajectory beginning at 23.
- At Layer 2, one isolates the deterministic valuation block beginning from the representation $23 = 2^3 \cdot 3 - 1$.
- At Layer 3, that block is recorded exactly by the BlockEntry coordinates $(u, m) = (3, 3)$.
- At Layer 4, the same block is represented only by the reduced structural state $(\omega, d) = (1, 4)$.

This example is intentionally small. Its purpose is only to show how one and the same trajectory fragment is successively reorganized, first into exact block coordinates and then into a reduced quotient state.

3. Definitions and Notation

This section introduces the objects used throughout the note in the same order as the layered reduction described above. The aim is to define each level of structure clearly before the induced map on reduced states is studied.

3.1. The Collatz Map

Let T be the Collatz map on the positive integers, defined by

$$\begin{aligned} T(x) &= x/2 && \text{if } x \text{ is even,} \\ T(x) &= 3x + 1 && \text{if } x \text{ is odd.} \end{aligned}$$

The ordinary dynamics of the conjecture are obtained by iterating T step-by-step.

3.2. Extended Valuation Blocks

An extended valuation block is a contiguous stretch of the Collatz dynamics beginning at an odd value of the form

$$x = 2^m u - 1,$$

where u is odd and $m \geq 1$. Along such a block, repeated odd-to-odd steps along the block remove exactly one factor of 2 at each stage until the block reaches its odd exit. The exact deterministic evolution of these blocks is established later; here the term is introduced only as the basic compressed unit of the dynamics.

3.3. BlockEntry Coordinates

A BlockEntry is the pair (u, m) associated with an odd value x of the form

$$x = 2^m u - 1,$$

with u odd and $m \geq 1$. These coordinates record the odd seed u and the 2 -adic entry depth m at the beginning of the block.

Whenever convenient, the corresponding odd value x may be recovered from (u, m) by the same formula.

3.4. Structural Odd Core and 3-adic Content

For a BlockEntry seed u , write

$$u = 3^a \omega,$$

where $a = v_3(u)$ and 3 does not divide ω . The quantity a records the 3 -adic content of u , while ω is the structural odd core obtained after removing all factors of 3 from u .

3.5. Reduced Structural State

The reduced structural state associated with a BlockEntry (u, m) is the pair (ω, d) , where

$$d = m + a.$$

Thus the reduced state retains the structural odd core ω together with the full combined depth parameter d , while no longer distinguishing between different decompositions of that total depth into separate contributions from m and a .

3.6. The Projection Q

Define the projection Q from BlockEntry coordinates to reduced structural states by

$$Q(u, m) = (\omega, d),$$

where $u = 3^a \omega$ with 3 not dividing ω and $d = m + a$.

This projection passes from exact block coordinates to the reduced state space studied in the rest of the note.

3.7. Induced Reduced Dynamics

The structural step on BlockEntry coordinates determines a corresponding transition on reduced states by passing through the projection Q . At this stage, the reduced map is introduced only at the level of construction:

- start from a BlockEntry (u, m) ,
- compute its structural-step output,
- project the result back to a reduced structural state.

The fact that this induced transition depends only on the reduced state (ω, d) , and not on the particular BlockEntry representative chosen, is a substantive result and is proved later in Section 8.

4. Dictionary of Symbols

This section collects the main symbols used throughout the note for quick reference. The formal definitions remain those of Section 3.

4.1. Layer 1: Raw Collatz Iteration

- T : the Collatz map on the positive integers.

- x : an integer in the raw Collatz dynamics; when discussing a block entry, typically an odd value of the form $x = 2^m u - 1$.

4.2. Layer 2: Extended Valuation Blocks

- **extended valuation block**: a deterministic block segment of the dynamics beginning from an odd value of the form $x = 2^m u - 1$ and continuing until its odd exit.

4.3. Layer 3: Exact Block Coordinates

- u : the odd seed in the BlockEntry coordinates.
- m : the 2-adic entry depth in the BlockEntry coordinates; equivalently, $m = v_2(x + 1)$.
- (u, m) : the BlockEntry coordinates associated with an odd value $x = 2^m u - 1$.
- E : the structural step on BlockEntry coordinates.

4.4. Layer 4: Reduced Structural States

- a : the 3-adic content of u , defined by $a = v_3(u)$.
- ω : the structural odd core of u , defined by $u = 3^a \omega$ with 3 not dividing ω .
- d : the full combined depth parameter, defined by $d = m + a$.
- (ω, d) : the reduced structural state associated with a BlockEntry.
- Q : the projection from BlockEntry coordinates to reduced structural states, given by $Q(u, m) = (\omega, d)$.
- F : the induced reduced map obtained by passing the structural step through the projection Q .

4.5. Symbols for a Single Structural Step

- A : the structural-step numerator $A = 3^d \omega - 1$.
- s : the 2-adic valuation of A , namely $s = v_2(A)$.
- x_{exit} : the odd exit of the structural step, given by $x_{\text{exit}} = A / 2^s$.
- m_+ : the next-entry depth, defined by $m_+ = v_2(x_{\text{exit}} + 1)$.
- u_+ : the next odd seed, defined by $u_+ = (x_{\text{exit}} + 1) / 2^{m_+}$.
- a_+ : the 3-adic content of u_+ , defined by $a_+ = v_3(u_+)$.
- ω_+ : the structural odd core of u_+ , defined by $u_+ = 3^{a_+} \omega_+$ with 3 not dividing ω_+ .
- d_+ : the next reduced depth parameter, defined by $d_+ = m_+ + a_+$ when the next BlockEntry is formed.

When no ambiguity arises, the subscripts $+$ indicate the corresponding quantities attached to the next block entry or next reduced state produced by a single structural step.

5. Transformations Between Representations

This section collects the main coordinate transforms used throughout the note. They are not new dynamical laws; rather, they describe how the same object is expressed at different layers of the formalism.

A useful schematic is:

$$x \rightarrow (u, m) \rightarrow (\omega, d) \rightarrow x_{\text{exit}} \rightarrow (u_+, m_+) \rightarrow (\omega_+, d_+).$$

5.1. Transform $X \rightarrow B$: Odd Integer to BlockEntry

Given an odd integer x , define

$$\begin{aligned} m &= v_2(x + 1), \\ u &= (x + 1) / 2^m. \end{aligned}$$

Then u is odd and

$$x = 2^m u - 1.$$

Accordingly, the associated BlockEntry is

$$B(x) = (u, m).$$

5.2. Transform $B \rightarrow R$: BlockEntry to Reduced State

Given a BlockEntry (u, m) , write

$$u = 3^a \omega,$$

where $a = v_3(u)$ and 3 does not divide ω . Then define

$$d = m + a.$$

Accordingly, the associated reduced structural state is

$$R(u, m) = (\omega, d).$$

This is exactly the projection Q introduced earlier.

5.3. Direct Transform $X \rightarrow R$: Odd Integer to Reduced State

Composing the previous two transforms gives a direct passage from an odd integer x to its reduced structural state.

Given odd x , define

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m = v_2(x + 1),  
u = (x + 1) / 2^m,  
a = v_3(u),  
ω = u / 3^a,  
d = m + a.
```

Then the associated reduced state is

```
R(x) = (ω, d).
```

5.4. Transform $R \rightarrow X_{\text{exit}}$: Reduced State to Structural Exit

Given a reduced structural state (ω, d) , define

```
A = 3^d ω - 1,  
s = v_2(A),  
x_exit = A / 2^s.
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Then x_{exit} is the odd exit of the structural step associated with (ω, d) .

5.5. Transform $X_{\text{exit}} \rightarrow R_+$: Exit to Next Reduced State

Given x_{exit} , define

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m_+ = v_2(x_exit + 1),  
u_+ = (x_exit + 1) / 2^{m_+},  
a_+ = v_3(u_+),  
ω_+ = u_+ / 3^{a_+},  
d_+ = m_+ + a_+.
```

Then the next reduced state is

$$R_+(x_{\text{exit}}) = (\omega_+, d_+).$$

5.6. Combined Structural Pipeline

Combining the previous transforms gives the full reduced structural pipeline:

$$x \rightarrow (u, m) \rightarrow (\omega, d) \rightarrow x_{\text{exit}} \rightarrow (u_+, m_+) \rightarrow (\omega_+, d_+).$$

This separates coordinate changes from the structural dynamics themselves. The map F acts on reduced states, but its construction may be understood as passing through these intermediate transforms.

Important distinction. The value x_{exit} is not another representative of the reduced state (ω, d) that produced it. The representative family $(u, m) = (3^r \omega, d - r)$ consists only of BlockEntry descriptions of the same initial reduced state. By contrast, x_{exit} is the odd value obtained after the structural step has been completed, and it serves as the input for constructing the next BlockEntry and next reduced state. Thus the transforms $X \rightarrow B \rightarrow R$ describe equivalent representations of a state, whereas $R \rightarrow X_{\text{exit}}$ is a genuine dynamical transition.

6. Determinism of Extended Valuation Blocks

This section establishes the exact algebraic behavior of an extended valuation block. The key point is that when an odd value is written in the form

$$x = 2^m 3^a \omega - 1,$$

with $m \geq 1$, $a \geq 0$, and ω odd with 3 not dividing ω , the next odd iterate preserves the same shape while decreasing the 2 -adic exponent by one and increasing the 3 -adic exponent by one. This is the mechanism underlying the deterministic block structure.

6.1. One-Step Preservation

Let

$$x = 2^m 3^a \omega - 1,$$

where $m \geq 1$, $a \geq 0$, and ω is odd with 3 not dividing ω . Then the next odd iterate of x is

$$T_{\{\text{odd}\}}(x) = (3x + 1) / 2.$$

Indeed,

$$3x + 1 = 3(2^m 3^a \omega - 1) + 1 = 2^m 3^{(a+1)} \omega - 2 = 2(2^{(m-1)} 3^{(a+1)} \omega - 1).$$

Therefore

$$T_{\text{odd}}(x) = (3x + 1) / 2 = 2^{(m-1)} 3^{(a+1)} \omega - 1.$$

Thus the same structural form is preserved under one odd step, with

$$m \rightarrow m - 1, a \rightarrow a + 1, \omega \rightarrow \omega.$$

In particular, exactly one factor of 2 is removed at this stage.

6.2. Deterministic Drift Along the Block

Iterating the one-step preservation identity yields the deterministic evolution of the entire block.

Lemma 6.2.1. Let

$$x_0 = 2^m 3^a \omega - 1,$$

with $m \geq 1$, $a \geq 0$, and ω odd with 3 not dividing ω . For each integer j with $0 \leq j \leq m - 1$, define x_j by repeated application of the odd-step map. Then

$$x_j + 1 = 2^{(m-j)} 3^{(a+j)} \omega.$$

Equivalently,

$$x_j = 2^{(m-j)} 3^{(a+j)} \omega - 1.$$

Proof. The case $j = 0$ is immediate from the definition of x_0 . If the formula holds for some $j < m - 1$, then

$$x_j = 2^{(m-j)} 3^{(a+j)} \omega - 1,$$

so by the one-step preservation identity,

$$x_{(j+1)} = 2^{(m-j-1)} 3^{(a+j+1)} \omega - 1.$$

This is exactly the required formula with j replaced by $j + 1$. The result follows by induction.

6.3. Endpoint of the Deterministic Block

At the final stage of the deterministic block, taking $j = m - 1$ gives

$$x_{(m-1)} + 1 = 2 \cdot 3^{(a+m-1)} \omega,$$

so

$$x_{(m-1)} = 2 \cdot 3^{(a+m-1)} \omega - 1.$$

Applying one further odd step gives

$$3x_{(m-1)} + 1 = 2(3^{(a+m)} \omega - 1).$$

Thus the odd exit of the block is obtained by dividing

$$3^{(a+m)} \omega - 1$$

by its full 2 -adic valuation.

6.4. Interpretation

An extended valuation block therefore evolves in a completely rigid way up to its odd exit: each deterministic odd-to-odd step replaces

$$x = 2^m 3^a \omega - 1$$

by

$$T_{\{\text{odd}\}}(x) = 2^{(m-1)} 3^{(a+1)} \omega - 1,$$

so one factor of 2 is removed from $x + 1$, one factor of 3 is gained, and the structural odd core ω remains unchanged.

This is why the entire block can be compressed exactly and why the quantity

$$3^{(a+m)} \omega - 1$$

naturally appears as the structural numerator governing the transition to the next block.

7. Construction of the Reduced Structural Step

Section 6 showed that an extended valuation block evolves deterministically up to its odd exit. The purpose of this section is to package that exact evolution into a single structural-step construction. This produces the bridge between BlockEntry coordinates and the reduced structural state formalism.

7.1. From BlockEntry Data to the Structural Numerator

Start with a BlockEntry (u, m) , and write

$$u = 3^a \omega,$$

where $a = v_3(u)$ and 3 does not divide ω . Then the corresponding odd block-entry value is

$$x = 2^m u - 1 = 2^m 3^a \omega - 1.$$

Applying Section 6 shows that the deterministic block evolves up to its endpoint according to the rigid drift formula, and that its odd exit is governed by the numerator

$$3^{(m+a)} \omega - 1.$$

Define the reduced structural depth by

$$d = m + a.$$

With this notation, the structural numerator becomes

$$A = 3^d \omega - 1.$$

This quantity is the structural numerator attached to the BlockEntry (u, m) , rewritten in reduced variables.

It is worth noting that the deterministic block evolution preserves the sum $m + a$: each odd-to-odd step decreases m by one and increases a by one. The reduced parameter d is exactly this preserved block-level invariant.

7.2. The Structural Step on BlockEntry Coordinates

Define the structural step E on BlockEntry coordinates as follows.

Given (u, m) :

1. write $u = 3^a \omega$ with $a = v_3(u)$ and 3 not dividing ω ,
2. define the reduced structural depth $d = m + a$,
3. form the structural numerator $A = 3^d \omega - 1$,
4. let $s = v_2(A)$,
5. define the odd exit $x_{\text{exit}} = A / 2^s$, which is odd by construction, since $s = v_2(A)$ removes the full 2 -adic valuation of A ,
6. define the next-entry depth $m_+ = v_2(x_{\text{exit}} + 1)$,
7. define the next odd seed $u_+ = (x_{\text{exit}} + 1) / 2^{m_+}$.

The resulting next BlockEntry is

$$E(u, m) = (u_+, m_+).$$

Thus a full deterministic block is compressed into a single transition from one BlockEntry to the next.

7.3. Passage to Reduced Structural States

Applying the projection Q to the output of the structural step produces the corresponding reduced state. That is, if

$$E(u, m) = (u_+, m_+),$$

then the next reduced structural state is obtained from

$$Q(u_+, m_+) = (\omega_+, d_+),$$

where $u_+ = 3^{a_+} \omega_+$ with 3 not dividing ω_+ and

$$d_+ = m_+ + a_+.$$

As noted in Section 5.6, x_{exit} is not another representative of the original reduced state (ω, d) , but the odd output of the structural step that initiates the next BlockEntry. That is, x_{exit} is part of the (ω_+, d_+) family.

This gives the candidate reduced transition associated with the original BlockEntry.

7.4. Interpretation

The structural step isolates the entire effect of one deterministic block in a single arithmetic object,

$$A = 3^d \omega - 1.$$

The quantity $s = v_2(A)$ measures the length of the 2 -adic cascade at the block exit, while the resulting odd value x_{exit} determines the entry depth and odd seed of the next block. In this way, the internal drift proved in Section 6 is converted into a block-to-block transition rule expressed through the reduced variables (ω, d) .

The next section shows that this construction depends only on the reduced state (ω, d) , and not on the particular BlockEntry representative chosen.

8. Well-definedness of the Reduced Map

Section 7 constructed the structural step on BlockEntry coordinates and showed how a corresponding reduced transition is obtained by projection. The purpose of this section is to prove that this reduced transition depends only on the reduced state (ω, d) , and not on the particular BlockEntry representative chosen. In other words, the reduced map is well-defined.

8.1. Representative-Independence

For a fixed reduced structural state (ω, d) , the BlockEntry representatives are precisely those pairs of the form

$$(u, m) = (3^r \omega, d - r),$$

where $0 \leq r \leq d - 1$. Indeed, if $Q(u, m) = (\omega, d)$, then writing $u = 3^a \omega$ with 3 not dividing ω gives $d = m + a$, so necessarily $u = 3^a \omega$ and $m = d - a$. Conversely, every such pair projects to the same reduced state (ω, d) .

Here r indexes the different BlockEntry representatives of the same reduced state. Increasing r shifts one unit of the total depth d from the entry-depth parameter m into the 3 -adic content of u , without changing the reduced state itself. Thus r measures movement within a representative family, not dynamical time evolution.

Thus, if two BlockEntry coordinates (u, m) and (u', m') satisfy

$$Q(u, m) = Q(u', m') = (\omega, d),$$

then they belong to the same representative family and may be written as

$$(u, m) = (3^a \omega, d - a), (u', m') = (3^{a'} \omega, d - a'),$$

for suitable integers $a, a' \geq 0$.

For every such representative, the structural numerator is the same:

$$3^m u - 1 = 3^{(d-a)} 3^a \omega - 1 = 3^d \omega - 1,$$

and likewise,

$$3^{(m')} u' - 1 = 3^{(d-a')} 3^{a'} \omega - 1 = 3^d \omega - 1.$$

Thus the quantity

$$A = 3^d \omega - 1$$

is determined entirely by the reduced state (ω, d) , independent of the chosen BlockEntry representative.

8.2. Consequences for the Structural Step

Since A depends only on (ω, d) , the same is true of its 2 -adic valuation

$$s = v_2(A),$$

and therefore also of the odd exit

$$x_{\text{exit}} = A / 2^s.$$

From x_{exit} , the next-entry depth

$$m_+ = v_2(x_{\text{exit}} + 1)$$

and the next odd seed

$$u_+ = (x_{\text{exit}} + 1) / 2^{(m_+)}$$

are likewise uniquely determined. Hence the resulting next BlockEntry

$$E(u, m) = (u_+, m_+),$$

once projected by Q , always produces the same reduced state regardless of which representative (u, m) with $Q(u, m) = (\omega, d)$ was chosen initially.

8.3. Main Proposition

Proposition 8.3.1. If two BlockEntry coordinates satisfy

$$Q(u, m) = Q(u', m'),$$

then

$$Q(E(u, m)) = Q(E(u', m')).$$

Proof. By Section 7.1, the reduced state (ω, d) uniquely determines the structural numerator $A = 3^d \omega - 1$. By Section 7.2, A uniquely determines s , x_{exit} , m_+ , and u_+ , and hence determines the projected output reduced state. Therefore the result of applying E and then projecting by Q depends only on (ω, d) , not on the representative chosen. This proves the claim.

8.4. The Induced Reduced Map

Because the projected output depends only on the reduced state, the induced reduced map F is well-defined. Accordingly, one may define

$$F(\omega, d) = Q(E(u, m))$$

for any BlockEntry representative (u, m) satisfying $Q(u, m) = (\omega, d)$.

This is the sense in which the reduced structural state space supports its own intrinsic dynamics: the distinctions discarded by the projection Q do not affect the next reduced state.

8.5. Interpretation

Section 6 showed how to compress a full deterministic block into a single structural step. Section 7 shows that this compression is compatible with the reduction from exact BlockEntry coordinates to reduced structural states. Equivalently, all BlockEntry representatives of the same reduced state produce the same next reduced state.

This is why the reduced dynamics may be studied directly on the state space of pairs (ω, d) , rather than only through the more detailed coordinates (u, m) .

9. Proved Structural Properties

With the reduced map now shown to be well-defined, one may ask which arithmetic features genuinely control the reduced dynamics. This section records the properties that follow directly from the structural-step construction and are established exactly. The purpose is to isolate what is already secure before turning later to empirical or conjectural patterns.

9.1. The Structural Numerator Governs the Step

For a reduced structural state (ω, d) , the corresponding structural step is governed by the quantity

$$A = 3^d \omega - 1.$$

By Sections 7 and 8, this numerator is determined entirely by the reduced state and is independent of the chosen BlockEntry representative. Its full 2-adic valuation

$$s = v_2(A)$$

therefore determines the odd exit

$$x_{\text{exit}} = A / 2^s,$$

and hence determines the next-entry depth and next odd seed. Thus the reduced transition is controlled by the arithmetic of A .

9.2. 2-adic Proximity to -1

The next-entry depth is measured by the 2-adic proximity of x_{exit} to -1 .

Lemma 9.2.1. For any odd integer y and any integer $n \geq 1$,

$$v_2(y + 1) \geq n \text{ if and only if } y \equiv -1 \pmod{2^n}.$$

Proof. By definition, $v_2(y + 1) \geq n$ exactly when 2^n divides $y + 1$. This is equivalent to $y + 1 \equiv 0 \pmod{2^n}$, or $y \equiv -1 \pmod{2^n}$.

In particular, if $y = x_{\text{exit}}$, then the next-entry depth

$$m_+ = v_2(x_{\text{exit}} + 1)$$

records exactly how close x_{exit} lies to -1 in the 2 -adic sense.

9.3. Parity of s Controls 3-gain

A central exact consequence of the reduced formalism is that the parity of s determines whether the next odd seed gains a factor of 3 .

Lemma 9.3.1. Let

$$A = 3^d \omega - 1,$$

with $s = v_2(A)$ and $x_{\text{exit}} = A / 2^s$. Then

$$3 \text{ divides } x_{\text{exit}} + 1 \text{ if and only if } s \text{ is even.}$$

Equivalently, the next odd seed u_+ acquires a factor of 3 precisely when s is even.

Proof. Since $A = 3^d \omega - 1$, one has $A \equiv -1 \pmod{3}$. Therefore

$$x_{\text{exit}} + 1 = A / 2^s + 1 \equiv -2^{(-s)} + 1 \pmod{3}.$$

Modulo 3 , one has $2 \equiv -1$, so $2^{(-s)} \equiv (-1)^s \pmod{3}$. Hence

$$x_{\text{exit}} + 1 \equiv 1 - (-1)^s \pmod{3}.$$

If s is even, this is congruent to $0 \pmod{3}$. If s is odd, it is congruent to $2 \pmod{3}$. This proves the claim.

Thus the parity of the 2 -adic cascade at the block exit determines whether the next stage gains additional 3 -adic content.

This matters because a 3 -gain changes how the next block is represented at the reduced level. When 3 divides $x_{\text{exit}} + 1$, the next odd seed u_+ carries additional 3 -adic content, so more of the next block-entry data is absorbed into a_+ rather than remaining in the structural odd core ω_+ . Equivalently, a 3 -gain affects how the next state decomposes into its reduced coordinates, and therefore influences the future evolution of both ω and d . In this sense, the parity of s does not merely record a congruence condition: it controls whether the next reduced state undergoes an additional structural absorption of a factor of 3 .

9.4. Special Case: $\omega = 1$

The family $\omega = 1$ provides a fully explicit test case for the reduced dynamics. Here

$$A = 3^d - 1.$$

Its 2 -adic valuation is given by the standard formula

$$\begin{aligned} v_2(3^n - 1) = \\ 1, & \text{ if } n \text{ is odd,} \\ 2 + v_2(n), & \text{ if } n \text{ is even.} \end{aligned}$$

Applying this with $n = d$ gives the following.

Proposition 9.4.1. If $\omega = 1$, then

$$s = v_2(3^d - 1)$$

satisfies

$$\begin{aligned} s &= 1 \text{ when } d \text{ is odd,} \\ s &= 2 + v_2(d) \text{ when } d \text{ is even.} \end{aligned}$$

In particular, for $\omega = 1$ the parity and size of s may be read off directly from d .

This family serves as a useful reference model because the reduced step is governed by an explicit valuation law rather than by a general mixed parameter pair (ω, d) .

9.5. What the Reduced Dynamics No Longer Need

For the purpose of computing the next reduced state, the separate values of the BlockEntry parameters $a = v_3(u)$ and $m = v_2(x+1)$ do not matter once ω and $d = m + a$ are fixed. Section 7 showed that different representatives in the same family

$$(u, m) = (3^r \omega, d - r)$$

all produce the same reduced output.

Accordingly, the reduced dynamics do not depend on the individual decomposition of d into a 3-adic contribution a and an entry-depth contribution m . What remains active is the reduced pair (ω, d) together with the valuation behavior of the associated structural numerator $A = 3^d \omega - 1$.

This identifies the exact level at which information has been compressed: the discarded distinctions do not affect the next reduced state, but the arithmetic of A still does.

10. Empirical Observations

This section records observations that appear repeatedly in computation and experimentation within the reduced structural framework, but which are not established here as theorems. They are included because they may help guide future reductions, suggest useful residue descriptions, or point toward stronger structural questions. Unless explicitly stated otherwise, the claims in this section should be read as empirical patterns or research heuristics rather than proved facts.

10.1. Residue Clustering in the Reduced Dynamics

Across computational samples, reduced states with the same or closely related residue data often appear to exhibit similar transition behavior. In particular, the valuation

$$s = v_2(3^d \omega - 1)$$

and the resulting 3-gain pattern seem to cluster by congruence classes of ω and by parity data attached to d .

At present, this is best regarded as a structural hint rather than a theorem. The reduced map is provably controlled by the arithmetic of $A = 3^d \omega - 1$, so it is plausible that residue-class organization should emerge; however, the extent to which finitely many congruence patterns govern the dynamics remains open.

10.2. Apparent Locking of s -parity in Some Families

In some experimentally explored families, the parity of

$$s = v_2(3^d \omega - 1)$$

appears to follow stable or repeating behavior over ranges of reduced states. Since Section 9.3 showed that the parity of s exactly controls whether the next stage gains a factor of 3 , any such parity regularity would have direct dynamical consequences.

At present, this should be treated only as an observed pattern. The note does not prove periodicity, eventual stability, or finite-state control of s -parity in general.

10.3. Near-Deterministic Average Behavior Within Some Residue Classes

Some residue classes appear empirically to produce reduced transitions with similar average behavior, even when individual steps still vary. This suggests that certain residue descriptions may organize the dynamics more rigidly than the raw pair (ω, d) alone makes obvious.

No exact mechanism for this apparent near-determinism is established here. In particular, the note does not prove that any bounded residue system captures the full reduced dynamics.

10.4. Possible Finite-State Shadow of the Reduced Map

The reduced formalism naturally raises the question of whether the dynamics admit a finite-state shadow: that is, whether some bounded collection of residue variables captures a large part of the observable step-to-step structure, while the unbounded behavior is carried mainly by the 2 -adic depth variables.

This possibility is motivated by computational behavior, not by proof. The current results establish only that the reduced map is governed by the arithmetic of $A = 3^d \omega - 1$ and by the valuations extracted from it. Whether this arithmetic can be organized into a genuinely useful finite-state model remains an open question.

10.5. The Role of the Approximate One-Third Phenomenon

Some computational patterns suggest that an approximate one-third-type behavior may emerge in families where the parity of s alternates or where 3 -gain occurs with a visible regularity. The significance of this is that repeated 3 -gain events do not simply mark local divisibility accidents: they repeatedly alter how much of the next step is absorbed into the reduced-coordinate decomposition, and so may influence longer-scale movement through the reduced state space. At present, this is only a heuristic interpretation of numerical patterns, not a proved statistical law.

If such behavior is real, it would likely need to be explained through the interaction between the 2 -adic cascade length s , the induced 3 -gain mechanism of Section 9.3, and the residue structure of ω and d . Establishing or refuting such a mechanism remains future work.

10.6. Status of These Observations

The observations above are included to mark where the reduced formalism appears most suggestive, not to enlarge the proved content of the note. They indicate directions where the reduced dynamics may admit further compression or classification, but they should not be interpreted as settled structural laws.

The next section records open questions suggested by this empirical picture.

11. Conjectures and Open Questions

The previous sections isolate the exact structural core of the reduced formalism and separate it from the empirical patterns that remain unresolved. This final section records the main mathematical questions suggested by the framework. Some are immediate structural questions about the reduced map itself; others concern whether the observed residue patterns reflect genuine arithmetic laws.

11.1. Classification of Reduced-State Transitions

The reduced map F is well-defined, but its effective structure is not yet classified. A central question is whether the transition

$$(\omega, d) \mapsto (\omega_+, d_+)$$

admits a cleaner direct description than the current step-by-step extraction through A , s , x_{exit} , and the next BlockEntry data.

In particular:

- Can $d_+ - d$ be expressed directly in terms of a small set of arithmetic parameters attached to $A = 3^d \omega - 1$?
- Can ω_+ be described through a more transparent residue law?
- Is there a more intrinsic formulation of F that avoids passing back through the full BlockEntry reconstruction?

11.2. Structure of the Valuation $s = v_2(3^d \omega - 1)$

The quantity s controls the 2 -adic cascade at the block exit and, through its parity, controls whether 3 -gain occurs. Understanding the reduced dynamics therefore requires a better description of the valuation sequence

$$v_2(3^n \omega - 1),$$

with $n = d$.

Open questions include:

- For fixed ω , what regularity does s exhibit as d varies?
- For fixed d , how is s organized by residue classes of ω ?
- Are there natural families for which the parity of s is periodic, eventually periodic, or rigidly constrained?
- Which parts of the observed behavior are consequences of classical 2 -adic lifting phenomena, and which are genuinely specific to this reduced formalism?

11.3. Mechanisms of 3-gain

Section 9 proved that the parity of s determines whether the next odd seed gains a factor of 3 . What remains open is to understand the broader structural role of repeated 3 -gain events.

Questions here include:

- How strongly does repeated 3 -gain influence long-term movement through the reduced state space?
- Can 3 -gain events be predicted efficiently from bounded residue data?
- Do repeated 3 -gain patterns explain any of the observed clustering or approximate average behavior in the reduced dynamics?

11.4. Representative Families and Further Compression

The well-definedness result shows that all BlockEntry representatives of a fixed reduced state produce the same reduced output. This raises a further structural question: is the current reduced state still carrying redundant information?

More specifically:

- Is the pair (ω, d) already minimal for describing the induced map, or does the repeated appearance of the invariant $m + a = d$ indicate that a cleaner parameterization exists?
- Can the notation and state description be compressed further without losing transparency?
- Is there a more natural coordinate system in which the structural numerator $A = 3^d \omega - 1$ appears even more directly?

11.5. Finite-State Shadow and Residue Control

The empirical observations suggest that residue data may organize much of the visible reduced dynamics. This leads to the question of whether the system admits a useful finite-state shadow.

Possible formulations include:

- Is there a bounded residue system that predicts the parity of s in large families?
- Can one isolate a finite collection of residue variables that explains most observed 3 -gain behavior?

- Does a finite-state model genuinely describe the reduced dynamics, or does the full arithmetic complexity reappear as soon as deeper 2 -adic behavior is tracked?

11.6. The Approximate One-Third Phenomenon

If the observed approximate one-third-type behavior reflects a real structural effect, then it should ultimately admit an arithmetic explanation inside the reduced formalism.

Open questions include:

- Is the phenomenon genuine, or only an artifact of limited computational samples?
- If genuine, does it arise from parity patterns of s , from frequencies of 3 -gain, or from a combination of both?
- Can it be formulated as a statistical statement on reduced-state transitions?

11.7. Relationship to Classical Collatz Formulations

The reduced formalism reorganizes the ordinary Collatz dynamics by compressing deterministic block structure and then quotienting by representative families. An important long-term question is how this framework relates to more classical odd-to-odd or accelerated Collatz maps.

In particular:

- Does the reduced map F recover known structures in a disguised form?
- Does it reveal genuinely new arithmetic organization not visible in standard formulations?
- Can insights from existing Collatz literature on valuations, accelerated maps, or residue graphs be translated cleanly into this reduced setting?

11.8. Closing Perspective

The framework developed in this note does not resolve the Collatz conjecture, but it does isolate a coherent reduced dynamical system whose exact structure can be stated cleanly. The main open problem is no longer whether the reduction is mathematically legitimate; that point has been established. The deeper challenge is to determine whether the reduced map F admits further compression, classification, or invariant structure beyond what is currently visible.

That question remains open, and it is the natural direction for any continuation of this program.